

Tsai-Shien Chen

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Education

University of California, Merced

Merced, CA

PH.D. CANDIDATE IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

Sep. 2022 - Now

- Advisor: Prof. Mukesh Singhal
- GPA: 4.00 / 4.00

National Taiwan University

Taipei, Taiwan

M.S. IN ELECTRONICS ENGINEERING

Sep. 2019 - March 2022

- Advisor: Prof. Shao-Yi Chien
- GPA: 4.30 / 4.30; Rank: 1st / 128

B.S. IN ELECTRICAL ENGINEERING

Sep. 2015 - June 2019

- GPA: 4.23 / 4.30; Rank: 5th / 190

Research Interests

- Image, Video, World Generative Models
- Video Dataset, Understanding, Generation, and Application Models
- Controllable and Personalized Content Creation

Honors & Awards

- 2026 **CVPR Outstanding Reviewers (top 5% of reviewers)**
- 2026 **CVPR Doctoral Consortium**
- 2025 **Google PhD Fellowship**
- 2024 **Graduate Student Opportunity Program Fellowship (one year of full tuition and stipend), UC Merced**
- 2023 **Best Master Thesis of the Year Award**, Graduate Institute of Electronics Engineering, NTU
- 2022 **Honorary Member (top 2% in college)**, Phi Tau Phi Scholastic Honor Society
- 2019 **Valedictorian**, Department of Electrical Engineering, NTU
- 2015-2019 **4-time Presidential Award (top 5% in department)**, NTU

Work Experiences

Adobe Research

San Francisco, CA

RESEARCH INTERN AT CREATIVE INTELLIGENCE LAB

March 2026 - Now

- Mentors: Richard Zhang, Eli Shechtman
- Autoregressive video streaming model

Snap Inc.

Santa Monica, CA

RESEARCH INTERN AT CREATIVE VISION TEAM

May 2023 - March 2026

- Mentors: Aliaksandr Siarohin, Sergey Tulyakov, Jun-Yan Zhu, Kfir Aberman
- Large-scale video dataset and video captioning, generation, and personalization models

Publications

Conference Proceedings

1. **Canvas-to-Image: Compositional Image Generation with Multimodal Controls**
Yusuf Dalva, Gordon Guocheng Qian, Maya Goldenberg, **Tsai-Shien Chen**, Kfir Aberman, Sergey Tulyakov, Pinar Yanardag, Kuan-Chieh Jackson Wang
SIGGRAPH 2026
2. **Omni-Attribute: Open-vocabulary Attribute Encoder for Visual Concept Personalization**
Tsai-Shien Chen, Aliaksandr Siarohin, Gordon Guocheng Qian, Kuan-Chieh Jackson Wang, Egor Nemchinov, Moayed Haji-Ali, Riza Alp Guler, Willi Menapace, Ivan Skorokhodov, Anil Kag, Jun-Yan Zhu, Sergey Tulyakov
CVPR 2026

3. **AlcheMinT: Fine-grained Temporal Control for Multi-Reference Consistent Video Generation**
Sharath Girish, Viacheslav Ivanov, **Tsai-Shien Chen**, Hao Chen, Aliaksandr Siarohin, Sergey Tulyakov
CVPR 2026
4. **Multi-subject Open-set Personalization in Video Generation**
Tsai-Shien Chen, Aliaksandr Siarohin, Willi Menapace, Yuwei Fang, Kwot Sin Lee, Ivan Skorokhodov, Kfir Aberman, Jun-Yan Zhu, Ming-Hsuan Yang, Sergey Tulyakov
CVPR 2025
5. **VIMI: Grounding Video Generation through Multi-modal Instruction**
Yuwei Fang, Willi Menapace, Aliaksandr Siarohin, **Tsai-Shien Chen**, Kuan-Chieh Jackson Wang, Ivan Skorokhodov, Graham Neubig, Sergey Tulyakov
EMNLP 2024
6. **Panda-70M: Captioning 70M Videos with Multiple Cross-Modality Teachers**
Tsai-Shien Chen, Aliaksandr Siarohin, Willi Menapace, Ekaterina Deyneka, Hsiang-wei Chao, Byung Eun Jeon, Yuwei Fang, Hsin-Ying Lee, Jian Ren, Ming-Hsuan Yang, Sergey Tulyakov
CVPR 2024
7. **Snap Video: Scaled Spatiotemporal Transformers for Text-to-Video Synthesis**
Willi Menapace, Aliaksandr Siarohin, Ivan Skorokhodov, Ekaterina Deyneka, **Tsai-Shien Chen**, Anil Kag, Yuwei Fang, Aleksei Stoliar, Elisa Ricci, Jian Ren, Sergey Tulyakov
CVPR 2024 (Highlight Presentation)
8. **Incremental False Negative Detection for Contrastive Learning**
Tsai-Shien Chen, Wei-Chih Hung, Hung-Yu Tseng, Shao-Yi Chien, Ming-Hsuan Yang
ICLR 2022
9. **Hard Samples Rectification for Unsupervised Cross-domain Person Re-identification**
Chih-Ting Liu, Man-Yu Lee, **Tsai-Shien Chen**, Shao-Yi Chien
ICIP 2021
10. **Tracklet-Refined Multi-Camera Tracking Based on Balanced Cross-Domain Re-Identification for Vehicles**
Kai-Siang Yang, Yu-Kai Chen, **Tsai-Shien Chen**, Chih-Ting Liu, Shao-Yi Chien
CVPR Workshops 2021
11. **Orientation-aware Vehicle Re-identification with Semantics-guided Part Attention Network**
Tsai-Shien Chen, Chih-Ting Liu, Chih-Wei Wu, Shao-Yi Chien
ECCV 2020 (Oral Presentation)
12. **Viewpoint-Aware Channel-Wise Attentive Network for Vehicle Re-Identification**
Tsai-Shien Chen, Man-Yu Lee, Chih-Ting Liu, Shao-Yi Chien
CVPR Workshops 2020
13. **Supervised Joint Domain Learning for Vehicle Re-Identification**
Chih-Ting Liu, Man-Yu Lee, Chih-Wei Wu, Bo-Ying Chen, **Tsai-Shien Chen**, Yao-Ting Hsu, Shao-Yi Chien
CVPR Workshops 2019

Pre-prints

1. **CineOrchestra: Unified Entity-Centric Conditioning for Cinematic Video Generation**
Sharath Girish*, **Tsai-Shien Chen***, Zhikang Dong, Mukesh Singhal, Hao Chen, Sergey Tulyakov, Aliaksandr Siarohin
Coming Soon
2. **LayerComposer: Multi-Human Personalized Generation via Layered Canvas**
Gordon Guocheng Qian, Ruihang Zhang, **Tsai-Shien Chen**, Yusuf Dalva, Anujraaj Argo Goyal, Willi Menapace, Ivan Skorokhodov, Meng Dong, Arpit Sahni, Daniil Ostashev, Ju Hu, Sergey Tulyakov, Kuan-Chieh Jackson Wang
arXiv preprint 2025

Patents

1. **Captioning Videos with Multiple Cross-Modality Teachers**
Tsai-Shien Chen, Yuwei Fang, Hsin-Ying Lee, Jian Ren, Aliaksandr Siarohin, Sergey Tulyakov
US12489956B2
2. **Scaled Spatiotemporal Transformers for Text-to-Video Synthesis**
Tsai-Shien Chen, Yuwei Fang, Anil Kag, Willi Menapace, Jian Ren, Aliaksandr Siarohin, Ivan Skorokhodov, Sergey Tulyakov
US20250245897A1

Professional Activities

Invited Talks	- ByteDance (April 2026): “From Data to Director: Towards Controllable and Personalized Movie Generation” - AI Time (Dec 2025): “From Personalization Models towards Movie Generator” - Anuttacon (July 2025): “Learning Video Generation and Personalization from Large-Scale Datasets”
Program Committee	- AAAI: 2026
Conference Reviewer	- CVPR’22,’23,’24,’25,’26; ICCV’21,’23,’25; ECCV’22,’24,’26; WACV’26; ACCV’24 - NeurIPS’23,’24,’25,’26; ICLR’24,’25,’26; ICML’24,’25,’26 - SIGGRAPH’26; SIGGRAPH Asia’26; ACMMM’25,’26
Journal Reviewer	- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) - IEEE Transactions on Image Processing (TIP) - International Journal of Computer Vision (IJCV)
Teaching Assistant	- UC Merced: CSE-185 Introduction to Computer Vision (Spring 2023) - UC Merced: CSE-024 Advanced Programming (Fall 2022) - NTU: EEE-5053 Computer Vision (Spring 2021)